

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

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## A Test of Mean Resolution Time

AP Statistics · Unit 4 · Topic 4.5 · B: 2027-01-21 · E: 2027-03-08

### [STAR] TODAY'S PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used  $\alpha = 0.05$ .

**Leah:** “The lower-tail test gives  $t = -1.5747$ ,  $df = 23$ , and  $p \approx 0.0645$ . I would state what evidence this does and does not provide.”

**Damon:** “Because  $p > 0.05$ , accept  $H_0$ . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

| Summary    | Value                  |
|------------|------------------------|
| Population | 1200                   |
| $n$        | 24                     |
| Mean (h)   | 11.1                   |
| SD (h)     | 2.8                    |
| Shape      | Symmetric; no outliers |

### FIRST TAKE — before the video (5 min)

Which account is more defensible? Cite the p-value comparison or a claim that a test can or cannot justify.

### [BOOK] COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)

For a one-sample mean test with unknown population standard deviation,  $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$  has  $df = n - 1$ . The  $p$ -value is the probability, assuming  $H_0$  is true, of a test statistic at least as extreme as the observed statistic in the direction of  $H_a$ . If  $p \leq \alpha$ , reject  $H_0$ ; if  $p > \alpha$ , fail to reject  $H_0$ . Failing to reject says the sample does not provide convincing evidence for  $H_a$  at that significance level. It does not prove the null value, and the conclusion must concern the population represented by the random sample.

**Finish after the video:**

**DOK 1** (a) Define the population mean and state hypotheses for the research question.

**DOK 2** (b) Verify random, 10 percent, and shape conditions. Compute  $SE$ ,  $t$ ,  $df$ , and the lower-tail p-value; interpret the p-value.

**DOK 3** (c) Adjudicate the accounts. Compare  $p$  with  $\alpha$ , distinguish the formal decision from acceptance or proof, state the scoped population conclusion, and explain the two beliefs produced by the rejected account.

**Frame:** The numerical comparison is  $p$  \_\_\_\_\_  $\alpha$ , so the formal decision is \_\_\_\_\_ and the population conclusion is \_\_\_\_\_.

**Frame:** That decision \_\_\_\_\_ establish \_\_\_\_\_ because the p-value is calculated assuming \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_